Apollo LM Fault Case Studies — Results Summary

Three fault studies, thirteen solves, at a glance

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Verdict

This vehicle fails through roll authority, not through thrust. The two faults that create a *thrust asymmetry* between the laterally-spaced engines ($y_{eng} = \pm 1.5$ m) push it past controllability while thrust-to-weight stays comfortable. The fault that creates no asymmetry is benign.

Study	Fault	Asymmetry?	Verdict
A Engine-out	one engine dead	total	**Infeasible** at design spacing. Needs $y_{eng} \leq 0.43$ m
B Degraded gimbal	one sluggish actuator	none	**Graceful.** All cases land; order 10–20 % in J
C Thrust efficiency	one engine delivers ηT	partial	**Unacceptable below** $\eta \approx 0.28$

All thirteen runs

Study	Case	Fault	J	Iters	Lands?
A	1	baseline, y_{eng}=1.5 m	2.8675e8	173	yes
A	2	engine 2 dead, y_{eng}=1.5 m	—	400	**NO**
A	3	engine 2 dead, y_{eng}=0.2 m	5.3298e8	190	yes
B	G1	baseline gimbal	2.8670e8	168	yes
B	G2	ω_n=1.5, ζ=0.70	3.5069e8	390	yes
B	G3	ω_n=0.6, ζ=0.70	3.1960e8	270	yes
B	G4	ω_n=1.0, ζ=0.25	3.4058e8	284	yes
C	E1	η=1.00 baseline	2.8670e8	168	yes
C	E2	η=0.85	2.9188e8	208	yes
C	E3	η=0.65	3.2113e8	199	yes
C	E4	η=0.40	3.7374e8	192	yes
C	E5	η=0.25	5.6255e8	204	**marginal***
C	E6	η=0.15	—	400	**NO**

* E5 converges but misses the landing gate: 5.95° touchdown attitude error against a 5° criterion, 0.193 m/s residual speed, and 74 s of the 80 s horizon consumed. Baseline gimbal is ω_n=4.0 rad/s, ζ=0.70. B/G1 and C/E1 are the same nominal problem and agree exactly — a useful cross-check.

The three numbers that matter

Boundary	Value	Meaning
$y_{crit} = d_{z,eng} \tan \delta_{y,\max} + L_{RCS}/T_{hover}$	**0.434 m**	max engine spacing for engine-out survivability (0.263 m on gimbal trim alone). Design value 1.5 m is **3.5× beyond**
$\eta_{sat} = (T_{hover}/2)/T_{\max,eng}$	**0.278**	below this the weak engine saturates and throttling can no longer equalise delivered thrust — **the practical limit**
η_{min}	**0.198**	below this the residual roll moment exceeds gimbal + RCS authority — infeasible

Each was derived analytically before solving, and each was confirmed by the solves. Study A predicted static trim would need 4.57° of yaw gimbal; the solve parked it at 4.65°. Study C predicted saturation at η_{sat}; at η=0.25 the weak engine's mean command was 22 101 N against a 22 520 N limit.

Why thrust is not the problem

Case	Total delivered thrust	T/W	Outcome
Nominal	45 040 N	3.59	lands
A, engine-out	22 520 N	1.80	**infeasible** (roll)
C, η=0.15	25 898 N	2.07	**infeasible** (roll)

Both failures retain ample thrust. What they lack is roll authority: the engine-out roll moment is **18 796 N · m** against **5 432 N · m** available (3 293 gimbal + 2 140 RCS) — a 13.4 kN · m deficit, which is 143 °/s² of roll acceleration against a 10 °/s rate limit, violated in 0.07 s.

What degradation costs when it is survivable

Quantity	Nominal	A, engine-out (y_{eng}=0.2)	B, worst gimbal	C, η=0.40
Time to contact	26 s	47 s	26 s	41 s
Peak total thrust	45 036 N	22 520 N	45 036 N	22 867 N
Mean yaw gimbal	1.2°	**4.65° of 6°**	1.5–1.6°	3.95°
RCS impulse	91 kN · s	171 kN · s (+88 %)	99 kN · s (+8 %)	147 kN · s (+62 %)
Wasted impulse	0	0	0	**909 kN · s (~40 %)**

Recurring pattern in both asymmetry faults: the dominant engine's yaw gimbal parks on a static-trim shelf near its limit, and RCS impulse rises sharply because the asymmetry must be held continuously rather than manoeuvred against.

Two findings that need a caveat

1. Study B's objective ordering is not reliable. The NLP is nonconvex. Three solves of the *identical* nominal problem under default multithreaded BLAS gave J spread over 1.0 %; pinning `OMP_NUM_THREADS=1` made it exact (2.867018e8, 168 iters, three times). But pinning does not make different *configurations* land in comparable minima — G2's objective moved **10.4 %** between two campaigns, comparable to the whole spread between cases. An earlier monotone ordering (+11.9/+12.6/+20.4 %) suggested "damping loss costs more than bandwidth loss"; the reproducible campaign (+22.3/+11.5/+18.8 %) does not support it, and that conclusion was **withdrawn**. Study B's robust findings are the physical ones (below). Study C's trend is safe — large, monotone, and corroborated by three independent physical measures. Study A does not rest on objectives at all.

2. Study B's compensation mechanism is counterintuitive but robust. The optimiser does *not* offload onto the healthy engine. It commands the degraded gimbal **harder** (RMS 1.51° → 2.70°) while the healthy engine's command *falls* (1.60° → 1.33–1.55°), because a sluggish actuator attenuates and must be over-driven. Max tracking error reaches exactly 12.000° — full scale, command at one ±6° limit while the deflection sits at the other. This held across both campaigns. Caveat: the optimiser exploits the actuator as a low-pass filter (only the healthy actuator's 4.04 rad/s bandwidth exceeds the 3.14 rad/s command grid Nyquist), issuing ±6° square waves no real autopilot would command — an artefact of penalising *commanded* rate rather than achieved deflection rate.

Caveats that matter for interpretation

- **Constant mass** — worst for Study C, where 1.34e6 N · s wasted at η=0.25 is ~440 kg of extra propellant on a 7 711 kg vehicle. A mass-tracking model might run out of propellant before roll authority. **Study C understates the propellant consequence.**
- **All faults known from** $t = 0$ — these answer "how well can a known-degraded vehicle be flown?", not "can it respond to a fault?"
- **No actuator slew-rate limit** — a real omission in Study B; a hard rate limit would hurt more than lost bandwidth and would block the over-commanding strategy.
- **Open loop, fixed 80 s horizon** — reference trajectories only. Studies A and C use 47 s and 74 s of the horizon, so it is closer to binding than nominal suggests.
- **Study A predates the thread-pinning protocol**, so its iteration counts are indicative; its verdicts are analytic and unaffected.

Design implication

The 1.5 m spacing is simultaneously the vehicle's best roll actuator (differential throttling gives ±4 000 N · m, roughly double the entire RCS roll budget) **and its single point of failure** — the same moment arm converts any thrust asymmetry into a trim problem it cannot absorb. Reducing the spacing relaxes both the engine-out and the η limits, at the cost of handing roll control to the propellant-limited RCS.

Study A's case 3 answers *"what spacing would have survived?"* — not *"can the vehicle as built survive?"* That answer is no.

Full detail, figures, and derivations: `case_studies_full_report.pdf`. Model documentation: `apollo_full_documentation.pd`
Per-study writeups sit in each case-study directory.